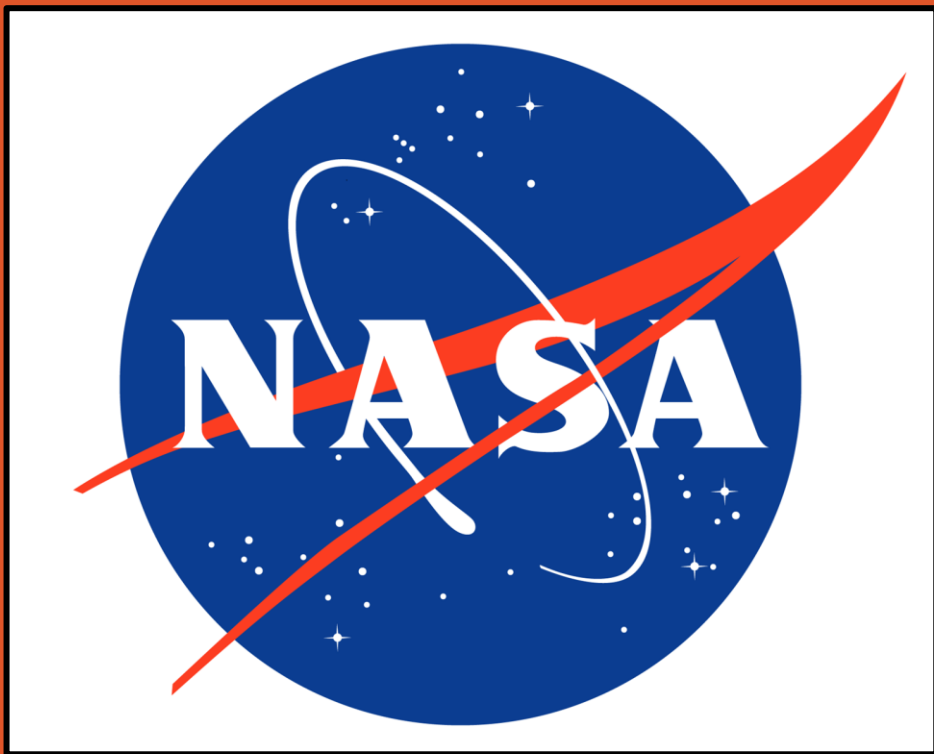




Background

- **Unprecedented Core Exploration:** NASA's Psyche mission targets a unique metal-rich asteroid believed to be an exposed planetary core, offering scientists a never-before-possible look at what lies beneath planetary crusts.
- **Potential Metallic Treasure:** Asteroid 16 Psyche may consist of up to 90% metal (primarily iron and nickel), representing potentially one of the most valuable celestial bodies in our solar system and a window into planetary formation.
- **Extreme Environment Challenges:** Engineers must overcome exceptional difficulties including minimal gravity (0.144 m/s²), dramatic temperature variations (1-200K), and the unprecedented metallic surface requiring innovative attachment solutions.

Fixture Design

- High-temp resistant materials, Titanium/Aluminum 6061 for structural components
- Specialized dual-action anchoring (electrostatic + thermite bonding)
- Designed for extreme thermal cycling (1-200K)
- System designed to handle height differences up to 3 feet and 34 degrees of tilt

Operational Phases

- **Transit Configuration:** Legs folded for compact storage during journey
- **Deployment Configuration:** Legs extend and prepare for surface contact
- **Anchored Configuration:** Two-stage process:
 - Initial: Electrostatic adhesion provides temporary secure hold
 - Permanent: Thermite welding creates final bond

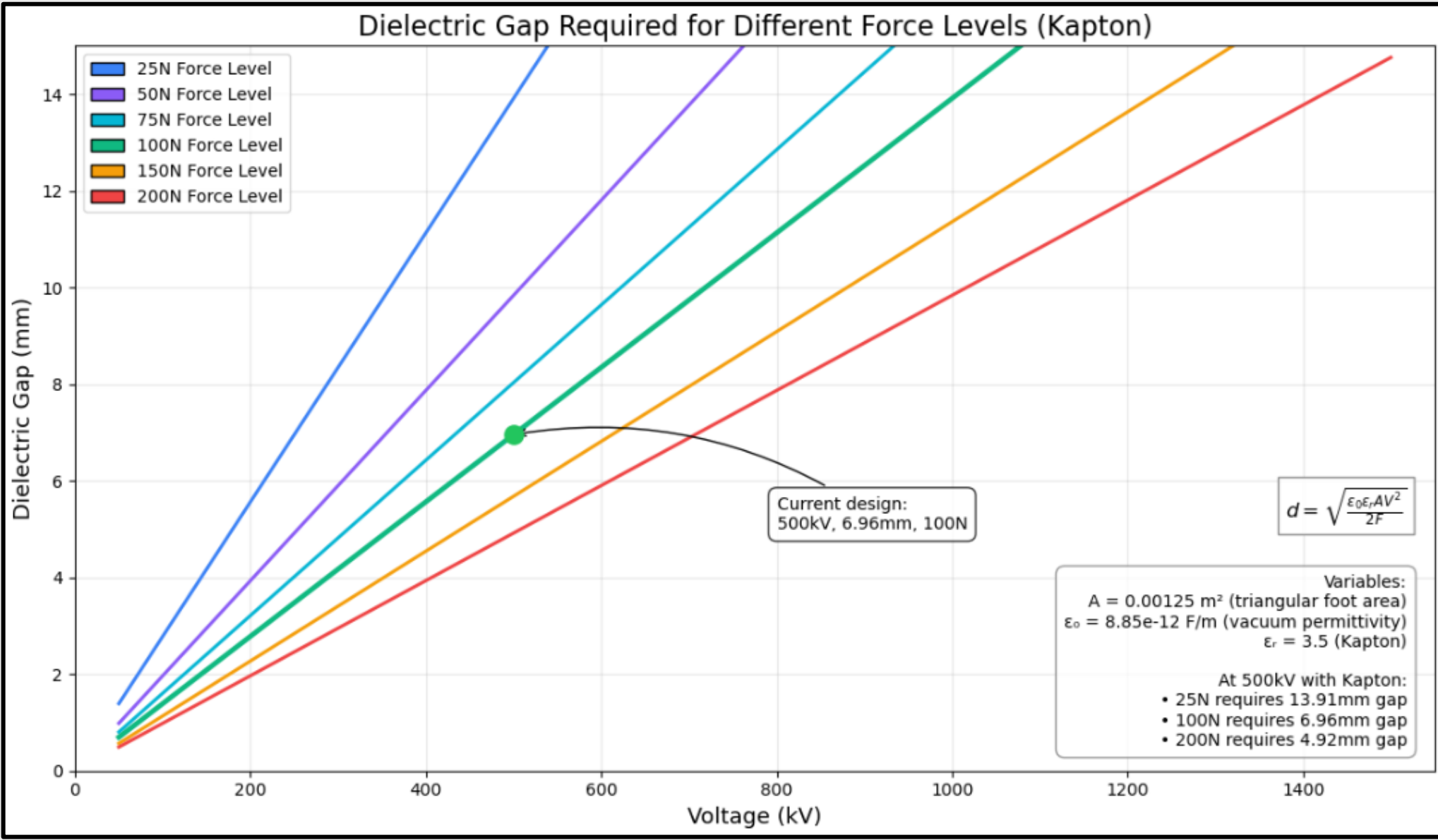
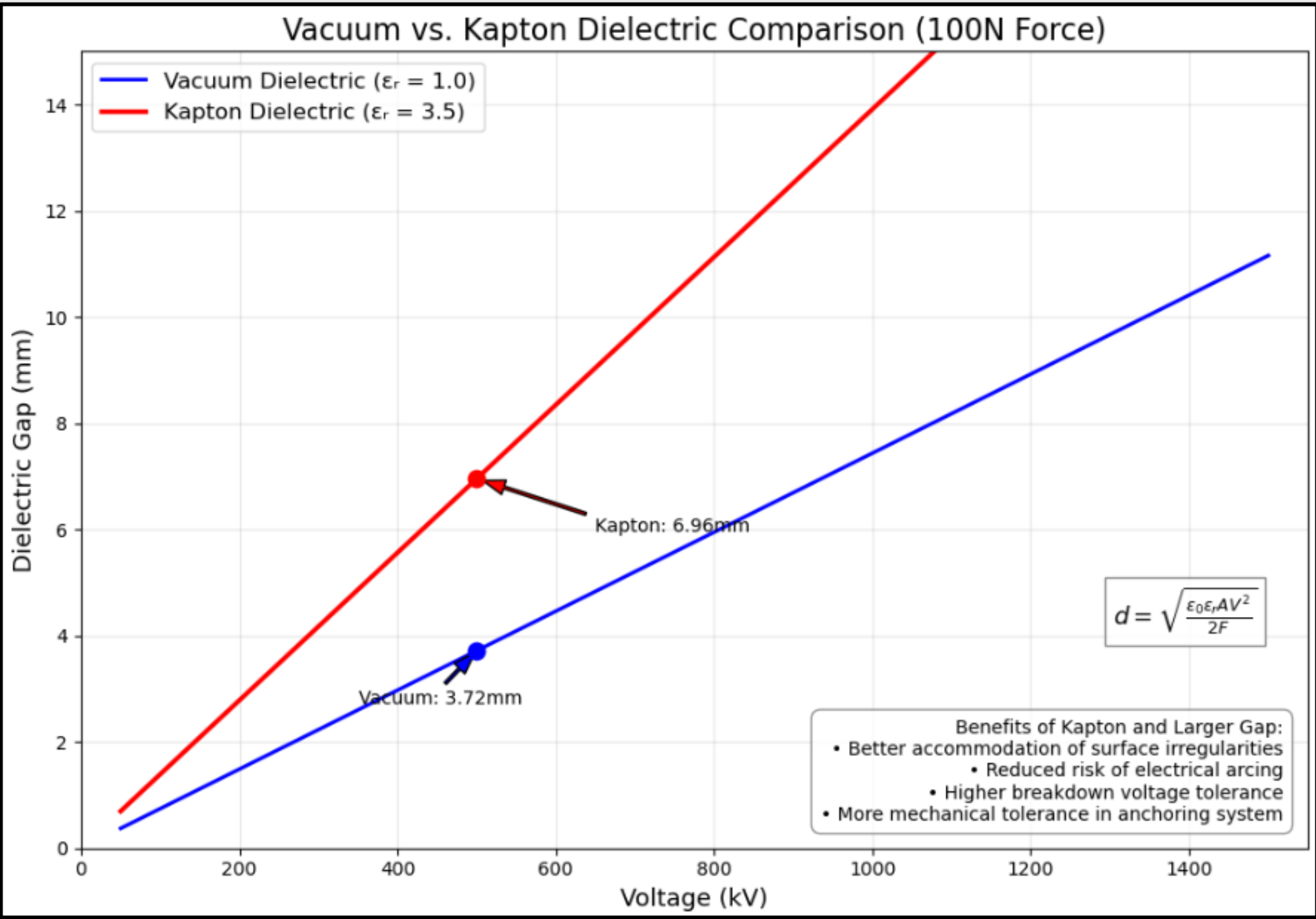
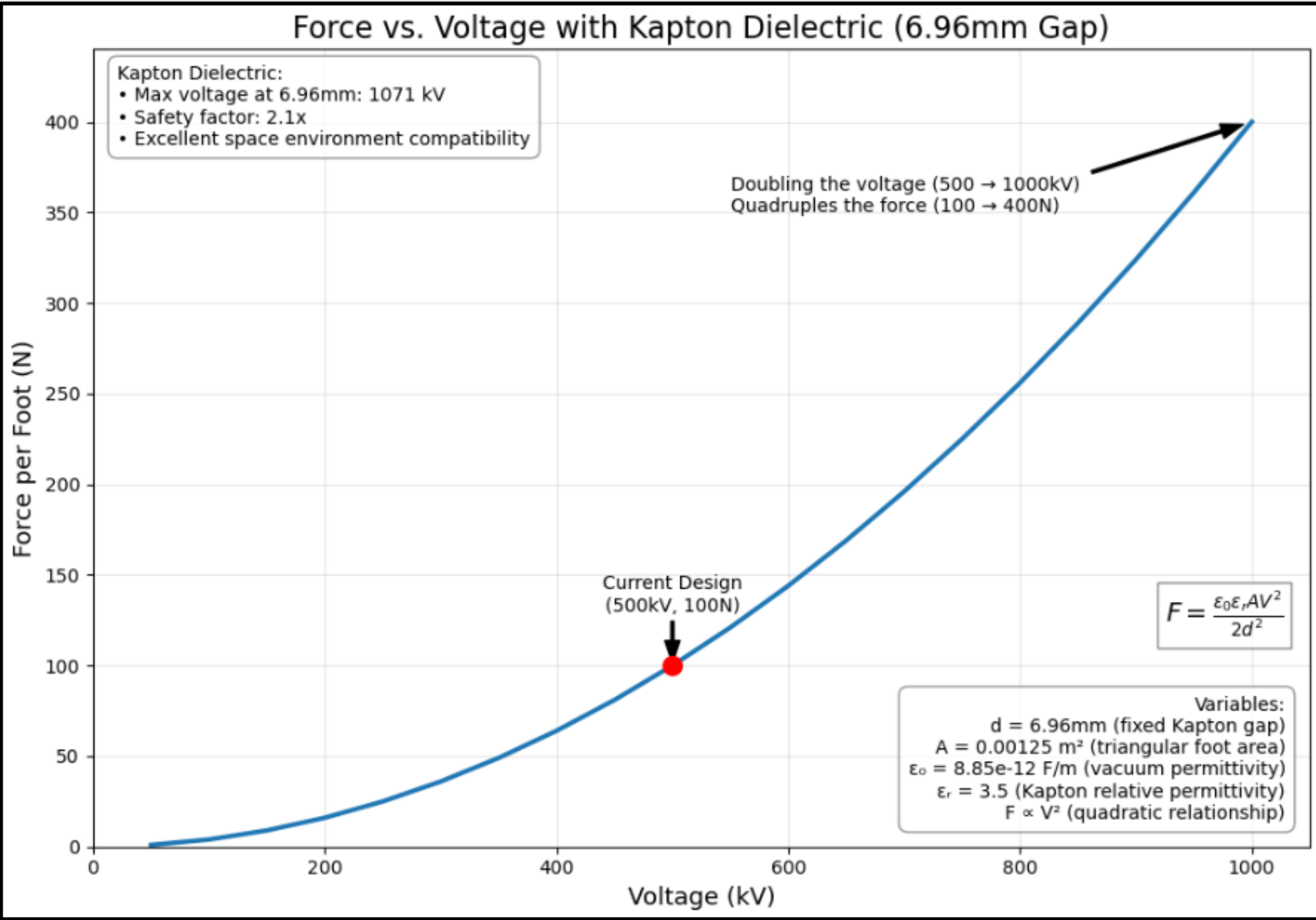


NASA Psyche
Space Elevator Anchor

Project Mission

Psyche is an asteroid with vast valuable resources, and we aim to develop an **anchor** for a space elevator tether to facilitate **sample transfer** between the asteroid and a geosynchronous orbiting spacecraft. Our team's primary goal is to create a well-designed anchoring system that can effectively secure to Psyche's unique ferrous surface in **microgravity conditions**. The system must adapt to **irregular terrain**, withstand **extreme temperature variations**, and provide sufficient **anchoring force** to support space elevator operations.

Electrostatic Anchor Design



System Features

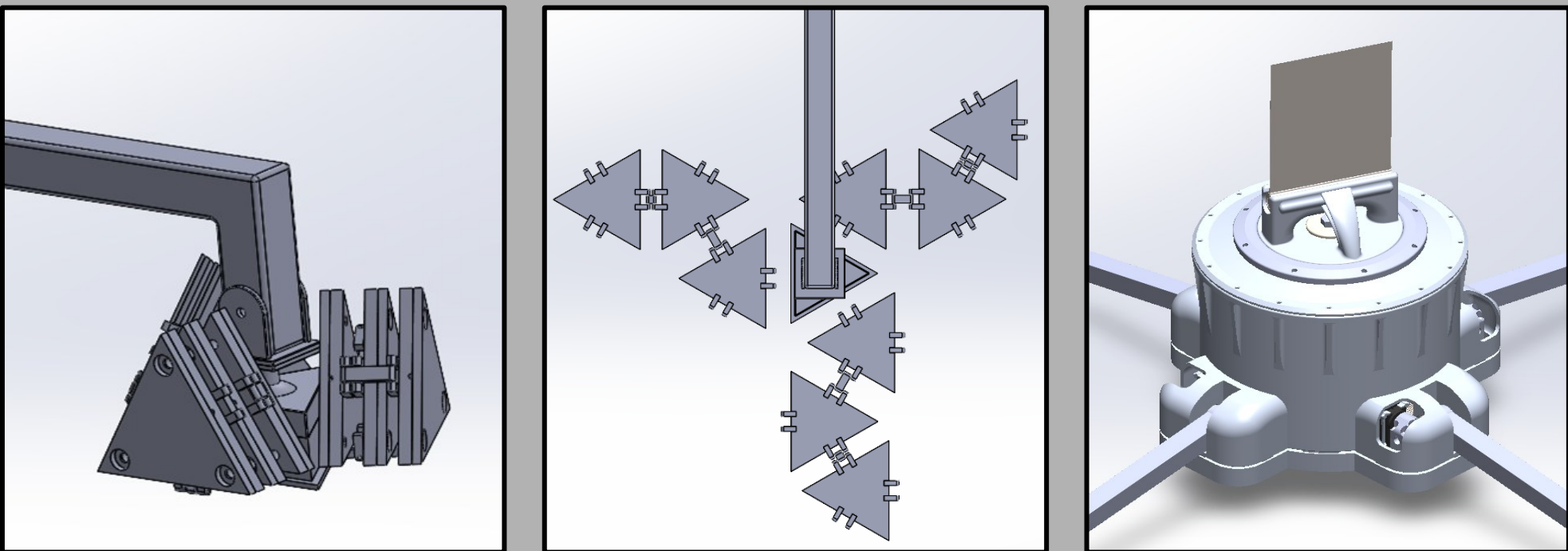
- **Central Mechanized Hub:** Houses motor and gear system with innovative clutch mechanism that independently controls each leg's position
- **Leg Modules:** Feature ball hinges and clutch mechanisms for flexibility and motion range to navigate irregular surface topography
- **Adaptable Feet Assembly:** Interconnected triangular segments that conform to Psyche's uneven metallic terrain, maximizing contact area
- **Tether Attachment Gimbal:** Flexible interface between anchoring system and space elevator ribbon

Primary References

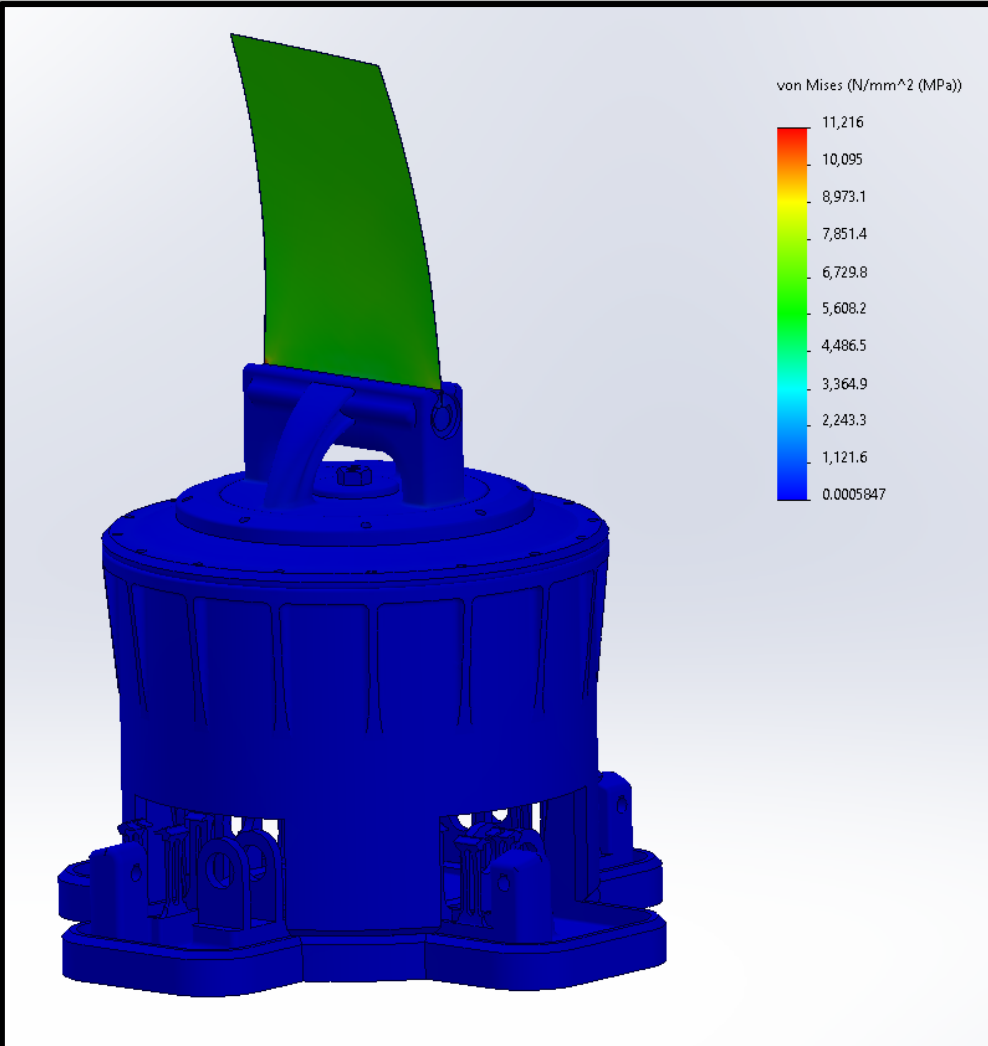
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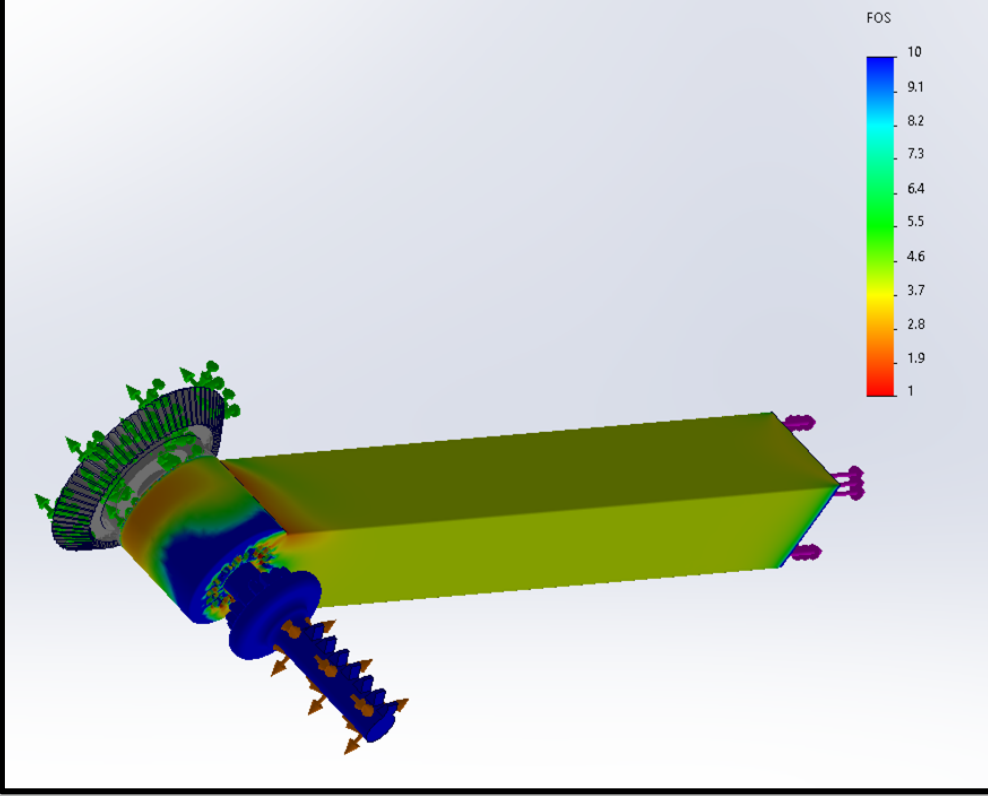
Preliminary Design



Finite Element Analysis

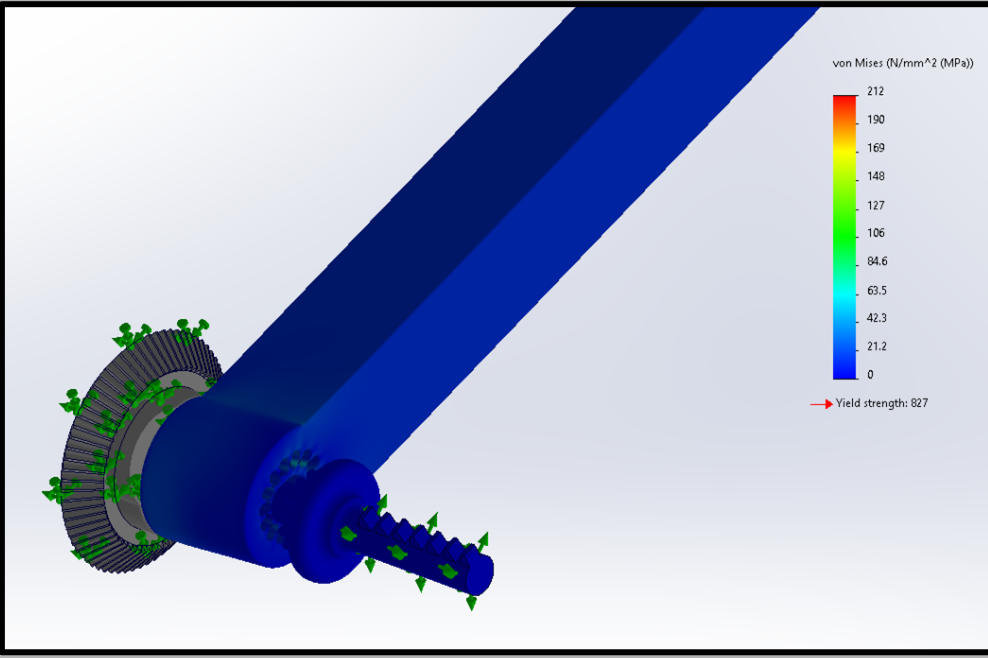


Central Mechanized Hub and Tether Attachment Gimbal stress distribution under 10 tons of upward tether force. The simulation visualizes von Mises stress concentrations (measured in MPa) across the device's structure with a FoS greater than 4. Tether simulated using titanium to approximate tensile strength similar to carbon composite material anticipated for the actual tether.



Fatigue Factor of Safety of Leg Axle Assembly showing 0.5 tons of Tensile Load Applied.

Areas where FoS < 2 implies future reiterations of design.



von Mises stress concentrations of Leg Axle Assembly showing 0.5 tons of Tensile Load Applied.

Leg stress does not exceed yield strength. Deformation is recoverable implying long cycle life.

Team Members

- **Project Manager:** Kyle Ledda-Lewaren
- **CAD/Design Lead:** JareI Shahim
- **Manufacturing Lead:** Kiefer Gonsalves
- **Robotics Lead:** Zachary Thomas
- **Faculty Advisor:** Sarah Oman
- **Sponsor:** NASA Psyche Mission

This work was created in partial fulfillment of Oregon State University Capstone Course MIME 498. The work is a result of the Psyche Student Collaborations component of NASA's Psyche Mission (<https://psyche.asu.edu>). "Psyche: A Journey to a Metal World" (Contract number NNM16AA09C) is part of the NASA Discovery Program mission to solar system targets. Trade names and trademarks of ASU and NASA are used in this work for identification only. Their usage does not constitute an official endorsement, either expressed or implied, by Arizona State University or National Aeronautics and Space Administration. The content is solely the responsibility of the authors and does not necessarily represent the official views of ASU or NASA.